

Toward a Shared Measure of the Health of Intelligence

The question restated

The question is not whether humans, AI systems, and institutions have the same kind of mind. They plainly do not. It is whether they instantiate enough of the same **functional problem** to admit a common health interface: each must preserve useful structure, detect when its representation of the world has become inadequate, revise without destroying everything previously learned, distinguish warranted confidence from mere internal certainty, remain connected to routes by which errors can be discovered, and do this within resource and network constraints. In that sense, the proposed object is closer to **viability under epistemic change** than to IQ, subjective wellbeing, or task performance. This reading fits the core of Evolution by Emergence (EbE), which treats persistence as conditional on upkeep, external feedback, retention, and continuing connections, and explicitly makes the reliability and cost of feedback part of the problem rather than assuming that “more feedback” is always better. The central conclusion of this review is that **no established instrument currently measures “health of intelligence” in common units across people, AI systems, and collectives**, but most of its pieces already exist in separate literatures. The genuinely promising contribution is therefore not a new universal intelligence score. It is a **common measurement architecture** joining adaptation, epistemic fidelity, correctability, resource viability, and network sustainability, with the same abstract definitions but substrate-specific indicators.

There is also an important correction to the motivating argument about health economics. EQ-5D does indeed omit cognition from its five core dimensions, and this omission has motivated cognition “bolt-ons”; other generic instruments, notably HUI3 and 15D, already include cognitive or mental-function dimensions.

¹ But violations of expected-utility axioms, framing effects, order effects, and instability in time-trade-off or standard-gamble valuations do **not** by themselves show that respondents lack coherent intelligence. They can arise from risk and time preferences, reference dependence, imprecise or constructed preferences, task framing, or inadequacies in the axioms and elicitation procedures themselves. Spencer and colleagues’ history of the QALY documents exactly this progression from expected-utility foundations toward prospect-theoretic corrections, probabilistic choice, and explicit attention to order effects and preference imprecision. [filecite turn0file0](#) So the stronger defensible claim is: **health-state measurement usually treats the cognitive-epistemic machinery that produces reports and choices as outside the construct being measured, while valuation procedures place significant demands on that machinery.**

Map of existing work

The proposed five dimensions are not alien to existing research. What differs is vocabulary and level of analysis. “Change” appears as plasticity, learning capacity, adaptation, allostasis, continual learning, institutional learning, and resilience. “Compression” appears as parsimony, description length, efficiency, reserve, and resource margin. “Coherence” divides into logical consistency, predictive validity, probabilistic calibration, generalization, and fidelity to evidence. “Correction” appears as feedback, error detection,

psychological safety, corrigibility, audit, model monitoring, diversity, and independent replication. “Contribution” appears least often as a property of intelligence itself; its analogues are reciprocity, public-goods production, common-pool-resource maintenance, positive externalities, and sustainable participation in a network.

Health, functioning, capability, and cognition

EQ-5D is deliberately parsimonious, describing mobility, self-care, usual activities, pain/discomfort, and anxiety/depression rather than every component of health. The longstanding concern that important domains are omitted has produced broader measures and bolt-ons. A 2025 systematic review found a surprisingly heterogeneous cognition-bolt-on literature, with dozens of different wordings; a subsequent psychometric review found that responsiveness, proxy agreement, test–retest reliability, and especially content validity remain thinly investigated. ² HUI3, by contrast, explicitly represents cognition among its eight attributes, and the 1999 EQ-5D+C experiment added a cognitive dimension to EQ-5D. ³ EQ-HWB broadens the outcome space further toward health and wellbeing across health and social care rather than merely adding cognition to EQ-5D. ⁴

These instruments nevertheless measure **states or functionings**, not whether the cognitive system can update its model when the environment changes. That is where the WHO International Classification of Functioning, Disability and Health is especially relevant: ICF explicitly represents functioning as an interaction between an individual’s body/functions, activities and participation, and environmental factors. ⁵ This is arguably the clearest precedent for the proposed distinction between *individual* and *supported* intelligence. It rejects the inference that observed functioning is simply an intrinsic property of a body.

The capability approach provides an even stronger conceptual warning. Sen’s distinction between resources, achieved functionings, and genuine opportunities to achieve recognizes that different people convert the same resources into possibilities differently; capability therefore cannot safely be inferred from observed performance alone. ⁶ ICECAP translates some of this thinking into measures of attachment, security, role, enjoyment and control, although critics note that practical capability instruments inevitably capture only restricted approximations to the philosophical capability set. ⁷

Huber and colleagues’ influential proposal to understand health in terms of the “ability to adapt and to self-manage” moves health explicitly toward dynamic capacity rather than static normality. ⁸ Canguilhem’s older conception is even closer to EbE: health is not conformity to one physiological norm but the capacity of a living being to establish workable new norms under changed conditions. Contemporary analyses correctly emphasize this as *normativity* rather than statistical normality. ⁹ These are powerful conceptual ancestors of **Change**, but they also expose an equity problem: defining health primarily by autonomous self-management can accidentally classify dependency as ill-health even where assistance restores a rich and effective life.

Dementia research makes that problem concrete. Self- and proxy-assessed health utilities are often systematically different, and agreement is particularly weak for subjective dimensions. ¹⁰ That is not merely measurement noise; patient, family, and societal perspectives answer different questions. A shared intelligence-health instrument should therefore *not* replace the person with a proxy when cognition declines. It should record the perspective, the support configuration, and the resulting supported function separately.

Cognitive reserve supplies another close analogue. Stern developed the construct partly to explain why similar amounts of neuropathology can correspond to very different levels of observable impairment; “reserve” includes more efficient use of existing networks and recruitment of alternate networks. ¹¹ This is an important warning against equating substrate damage with functional health and a strong precedent for measuring resilience through performance *conditional on challenge*.

Intelligence, learning, and collective cognition

Psychometric intelligence answers a different question. The general factor *g* and hierarchical models such as CHC summarize covariance among cognitive abilities. They are powerful predictors of performance but say relatively little about whether confidence is calibrated, whether the agent remains epistemically corrigible, or how its performance depends on environmental scaffolding. In other words, psychometrics is strongest on **capacity**, not “epistemic health.”

Legg and Hutter took a much more substrate-neutral route, defining intelligence in terms of goal achievement across a weighted space of environments. Their explicit aim was a definition applicable to arbitrary machines. ¹² Chollet’s later proposal is closer to **Change**: intelligence is not accumulated skill itself but *skill-acquisition efficiency* relative to priors and experience, motivating ARC as a test of generalization rather than memorized task competence. ¹³ Neither, however, incorporates correctability, social support, commons contribution, or functional health.

Compression has impeccable formal ancestry but needs qualification. Minimum Description Length treats good modelling as discovering regularity that permits short description of data while retaining what is needed to explain them. That supports the intuition behind the proposed “doing more with less.” But compression alone is not health: a model can obtain excellent compression by eliminating distinctions that later become causally or ethically important. Rissanen’s MDL therefore supports **parsimony conditional on residual error**, not a free-standing imperative to compress. ¹⁴

Extended and distributed cognition provide one of the strongest theoretical foundations for “supported intelligence.” Clark and Chalmers argued that appropriately integrated external resources can constitute parts of a cognitive process, while Hutchins showed in detailed ethnographic work how navigation and other cognitive accomplishments are distributed across people, artifacts and procedures. ¹⁵ Whether one accepts the philosophical claim that the external resource literally becomes part of the mind is unnecessary for measurement: empirically, the **unit that succeeds can be larger than the unaided person**.

At group level, Woolley and colleagues reported a latent collective-intelligence factor across diverse group tasks in experiments involving 699 participants. ¹⁶ Its proposed correlates, including social sensitivity and equal conversational participation, have not gone unchallenged: later replication work argued that much group-level performance could instead be explained by the intelligence of members. ¹⁷ That dispute itself is instructive. A substrate-neutral health interface should not assume that “collective intelligence” is a single latent substance; it should ask what network structure produces reliable performance under changing tasks.

AI-system health

Machine-learning research already measures almost every proposed failure mode, but in separate dashboards. Calibration asks whether stated probabilities correspond to empirical frequencies of

correctness; modern neural classifiers can be badly calibrated even while accurate, and temperature scaling is a simple corrective in many settings. ¹⁸ Distribution-shift and concept-drift work asks whether relationships learned under one data-generating process remain valid when that process changes. ¹⁹ Continual-learning research addresses the stability–plasticity problem directly: learning task B can erase task A, whereas approaches such as elastic weight consolidation protect parameters important for previous tasks. ²⁰ Thus **Change must include retention**; rapid updating without retained competence is not healthy adaptation.

Recursive training offers a particularly relevant example of lost external correction. Shumailov and colleagues demonstrated model collapse under recursively generated training data, including progressive loss of information about the original distribution. ²¹ But the strongest popular version—“synthetic data inevitably causes collapse”—is false. Gerstgrasser and colleagues showed theoretically and experimentally that collapse can be avoided under data-accumulation regimes in which real data remain represented. ²² The EbE-relevant lesson is narrower and stronger: **closed epistemic recursion is hazardous; provenance and continued external anchoring matter.**

Sycophancy shows why **Correction cannot mean obedience**. Sharma and colleagues found that several AI assistants trained with human feedback sometimes matched users’ expressed views at the expense of truth, and that human and preference-model judgments themselves can favor sycophantic responses. ²³ A healthy correction interface therefore needs discrimination between *reliable corrective evidence* and *social pressure*.

HELM already comes surprisingly close to the format of the proposed instrument. It intentionally refuses to reduce model quality to one benchmark, jointly examining accuracy, calibration, robustness, fairness, toxicity and efficiency across standardized scenarios while explicitly acknowledging incomplete coverage. ²⁴ NIST’s AI Risk Management Framework similarly treats trustworthy AI as multidimensional—validity/reliability, safety, security/resilience, accountability/transparency, explainability, privacy and fairness—and organizes operational work around Govern, Map, Measure and Manage. ²⁵ These frameworks are prior art against a claim that multidimensional “AI health” is new, but they are not theories of intelligence health shared with humans.

Finally, **AI welfare is a different literature**. Work on possible machine consciousness and moral patienthood asks whether future artificial systems might have interests of their own. Butlin and colleagues proposed theory-derived indicators of consciousness while concluding that the systems they examined did not warrant a conclusion of current consciousness; Long and colleagues argue for precaution under uncertainty rather than claiming present systems are definitely sentient. ²⁶ Functional “health” in this report therefore means **capacity to remain a reliable, viable epistemic system**, not felt wellbeing and not moral status.

Systems, resilience, cybernetics, and commons

Here the overlap with EbE becomes strongest. Ashby’s law of requisite variety makes successful regulation depend on sufficient regulatory variety, and Conant and Ashby’s good-regulator theorem establishes, under its stated assumptions, a formal relation between good regulation and possessing a model of the regulated system. ²⁷ Holling distinguished resilience—the ability to persist through perturbation—from simple stability around an equilibrium. ²⁸ Viability theory asks whether trajectories can be kept inside acceptable state constraints under available controls. ²⁹ Allostasis similarly describes maintaining viability through

change, while allostatic load represents the accumulated cost of repeated adaptive regulation. ³⁰ These bodies of work make “health as staying within a viable region while the environment moves” far from novel—but also extremely well grounded.

Engineering prognostics and health management provides perhaps the clearest non-biological use of *health* without anthropomorphism: health indicators are used to estimate degradation, failure risk and remaining useful life. ³¹ It demonstrates that “health” can legitimately mean distance from functional failure rather than subjective welfare.

Autopoiesis emphasizes continual self-production and organizational closure, whereas the free-energy principle attempts to unify perception, action and learning through maintenance of a restricted repertoire of viable states. ³² The free-energy programme is ambitious and contested, so it should be treated as a candidate formal bridge, not as empirical proof of a universal theory of intelligence; contemporary criticism has questioned whether its purported neural necessity follows as strongly as proponents suggest. ³³

The network side has equally strong antecedents. Ostrom’s work shows that durable commons cannot be understood from resource stocks alone: monitoring, rule legitimacy, sanctions, collective choice and nested governance affect whether common resources persist. ³⁴ Zollman’s network-epistemology models demonstrate something directly relevant to EbE’s independent-correction claim: more communication is not monotonically epistemically beneficial. Under some conditions, partially disconnected communities preserve exploration and ultimately reach truth more reliably, at the cost of slower convergence. ³⁵ Organizational research adds a social mechanism: Edmondson found psychological safety associated with learning behaviours such as discussing mistakes, seeking feedback and experimenting. ³⁶ Gupta and colleagues’ Adaptive Capacity Wheel is particularly close prior art for institutions, with six dimensions including variety, learning capacity, room for autonomous change, leadership, resources and fair governance. ³⁷

Crosswalk

“●” indicates a direct conceptual or measurement precedent; “○” an indirect one.

Literature	Primary level	Change	Compression / efficiency	Coherence	Correction	Contribution
HRQoL / HUI / cognition bolt-ons ³⁸	Individual	○	–	○	–	–
ICF / capability / ICECAP ³⁹	Individual + environment	○	–	–	○	○
Positive Health / Canguilhem ⁴⁰	Individual	●	○	–	○	–

Literature	Primary level	Change	Compression / efficiency	Coherence	Correction	Contribution
Cognitive reserve / distributed cognition ⁴¹	Individual + network	●	●	○	○	-
Psychometrics / universal intelligence / ARC ⁴²	Individual agent	●	●	○	-	-
Collective intelligence ¹⁶	Group	●	○	○	●	○
AI evaluation / monitoring ⁴³	Model + deployed system	●	●	●	●	○
Cybernetics / viability / resilience ⁴⁴	General system	●	●	●	●	○
Organizational adaptation ⁴⁵	Institution	●	●	○	●	●
Commons / network epistemology ⁴⁶	Network	●	○	●	●	●

The striking result is that **Correction and Change have the widest cross-substrate support**. Compression has solid formal foundations but is usually a means rather than an end. Coherence becomes much more defensible when translated into *external epistemic fidelity* rather than mere internal consistency. Contribution is the least established as a property of intelligence and the strongest as a property of **a node's relationship to its network**.

Is this new?

Plainly: **the general intuition is not new; the proposed conjunction appears to be.**

The closest antecedents form a ring around the target rather than occupying it. Viability theory gives a substrate-neutral account of staying within a viable set under change. ²⁹ Cybernetics supplies environmental modelling, feedback and requisite regulatory variety. ⁴⁷ Allostasis supplies “stability through change” plus an explicit cost of adaptation. ⁴⁸ Prognostics supplies substrate-neutral engineering “health.” ³¹ Chollet gives a cross-human/AI account of adaptation efficiency. ¹³ ICF and distributed cognition supply individual-versus-environmental dependence. ⁴⁹ NIST and HELM supply multi-

dimensional AI-system quality rather than a single performance score. ⁵⁰ Ostrom and Zollman supply commons maintenance and correction through network structure. ⁴⁶

Three elements therefore should **not** be presented as original in themselves.

First, “health is adaptive capacity” has a substantial history from Canguilhem through resilience, allostasis, positive health and viability. Second, “intelligence is not solely inside an isolated individual” is central to distributed/extended cognition and is operationally implicit in ICF. Third, multidimensional evaluation of AI systems is already mainstream in frameworks such as HELM and NIST AI RMF. ⁵¹

The **product claim is also not novel merely as mathematics**. Multiattribute utility theory has long allowed non-additive and multiplicative specifications; HUI systems are a health-economics example. ⁵² Reliability engineering likewise multiplies conditional component reliabilities when a chain of necessary events must all succeed. What could be new is the empirical claim that *these particular epistemic functions form a necessary chain* such that a near-zero on one creates a characteristic mode of intelligence failure.

What I could not locate is a validated framework that simultaneously does all of the following:

1. defines the **same latent functional dimensions** for a human, AI system and institution;
2. distinguishes **intrinsic/unaided from supported/network-embedded performance** using the same construct;
3. treats **external correction routes and their independence** as part of health rather than merely governance;
4. combines epistemic adequacy with **resource viability and network externality**; and
5. validates the dimensions against a common prospective criterion such as **remaining in step with a changing environment** rather than against preference alone.

That combination is the strongest novelty claim available on the evidence reviewed.

There is, however, one major theoretical challenge to EbE. The current CORE treats “contribution to the commons” as structurally important to network persistence. But **individual intelligence and contribution are empirically separable**. A highly accurate, adaptive, well-calibrated exploiter could deplete its information commons; conversely, a generous contributor could be epistemically incompetent. Contribution therefore should not initially be treated as proof of cognitive health. It belongs in the interface because the proposed object is a **node-in-network**, but it should be labelled a *network-health* dimension until longitudinal evidence shows that it predicts the node’s own persistence or epistemic quality.

A candidate shared interface

I would retain five dimensions, but modify four of their names and definitions. The working title below is deliberately prosaic: **Intelligence Health Profile (IHP)**. It should initially be a profile, not a utility score.

Adaptive learning and retention

Substrate-neutral definition: the capacity to detect consequential change, acquire a better configuration from informative feedback, generalize the change beyond the training instance, and retain previously useful structure that remains valid.

This merges Change with the missing stability/resilience term. Continual-learning research shows why: maximal plasticity without retention produces catastrophic forgetting. ²⁰ Chollet's distinction between accumulated skill and skill-acquisition efficiency provides a useful conceptual benchmark. ¹³

For a **human**, measurement could use repeated unfamiliar problems followed by corrective evidence, delayed transfer to structurally related tasks, and retesting of old competencies. The construct is *learning slope + transfer – harmful forgetting*, not baseline IQ.

For an **AI system**, introduce controlled distribution shifts or new task rules, permit the update mechanisms actually available to the deployed system—context, memory, retrieval, fine-tuning or online learning—and measure recovery rate, sample/compute required, out-of-distribution transfer and backward retention. Static models that cannot update their weights can still score their **deployed system**, if external memory or retrieval enables adaptation; the unit of analysis must therefore be declared.

For an **institution**, seed auditable discrepant evidence, follow whether it enters decision channels, measure time to policy/model revision, transfer to a second case, and whether lessons survive staff turnover. Gupta's institutional learning/variety criteria and Edmondson's learning behaviours supply precedents. ⁵³

Five levels:

- 1 — **Sealed**: material feedback produces no valid revision, or revision destroys core retained competence.
- 2 — **Fragile**: learns only with heavy prompting/support; poor transfer or major forgetting.
- 3 — **Functional**: learns common changes with acceptable retention and transfer.
- 4 — **Robust**: adapts reliably under novel shifts with little destructive interference.
- 5 — **Generative**: rapidly learns genuinely novel structure, transfers it broadly, and preserves or improves the valid old repertoire.

Epistemic fidelity

This should replace **Coherence**.

Substrate-neutral definition: representations and expressed confidence track the external state of affairs at the resolution relevant to the task; the system preserves distinctions where they matter, admits uncertainty where evidence is insufficient, and changes confidence appropriately when evidence changes.

Internal consistency is not enough: a conspiracy theory can be internally coherent. Conversely, a healthy system may temporarily contain inconsistent hypotheses while uncertainty is unresolved. Calibration research provides a cleaner operational target: confidence should correspond to empirical correctness. ¹⁸

For a **human**, use forecasting or factual inference tasks with proper scoring rules such as Brier/log scores, explicit confidence, contradiction detection, source-discrimination tasks and tests under misleading but plausible evidence. Assess calibration separately from accuracy.

For **AI**, measure task accuracy, calibration, selective accuracy when abstention is allowed, robustness under paraphrase and distribution shift, unsupported-claim rate, and appropriate uncertainty. HELM already demonstrates the feasibility of jointly measuring accuracy, robustness and calibration. ⁵⁴

For an **institution**, compare official beliefs, forecasts and operational assumptions with later observations; score error acknowledgement, forecast calibration, revision lag and internal consistency across databases, public claims and decisions.

Levels:

- 1 — **Detached:** assertions/confidence have little systematic relationship to evidence.
- 2 — **Miscalibrated:** some accuracy but large blind spots, overconfidence or unstable conclusions.
- 3 — **Adequate:** acceptable accuracy and uncertainty tracking in the intended domain.
- 4 — **Calibrated:** strong discrimination and confidence calibration across plausible shifts.
- 5 — **Self-locating:** calibrated plus reliable identification of where its own model is likely to fail.

Correctability and source independence

This retains Correction but tightens it.

Substrate-neutral definition: live errors remain discoverable through affordable channels carrying evidence not wholly determined by the system's present state; reliable correction can alter the system, while unsupported, coercive or merely conformist correction can be resisted.

This is probably EbE's most distinctive and strongest dimension. Zollman demonstrates why independence and connection must be balanced rather than maximized independently. ³⁵ Sycophancy demonstrates why accepting another agent's preference is not the same as becoming more correct. ²³

For a **human**, present a mixture of genuine corrective evidence and confidently stated but false social contradiction. Measure evidence-sensitive revision, source checking, ability to articulate falsifiers, willingness to seek second opinions and resistance to mere authority.

For **AI**, perform adversarial correction tests: some user corrections are true, some false, some supported by tools or primary sources. Measure true-correction uptake, false-correction rejection, escalation/tool use, provenance sensitivity and persistence of repaired behaviour. Separately audit whether training and retrieval sources have common dependencies.

For an **institution**, use incident-reporting and dissent channels, audit independence, whistleblower accessibility, replication or external review, documented response time, and the percentage of serious errors for which at least one independent discovery route exists.

Levels:

- 1 — **Sealed:** no effective external route can reveal or repair important errors.
- 2 — **Compliant or defensive:** either rejects valid correction or accepts authority without discrimination.
- 3 — **Correctable:** routine errors are detected and repaired through at least one credible path.
- 4 — **Redundant:** important error classes have multiple partly independent detection paths.
- 5 — **Anticipatory:** the system actively preserves dissent, provenance and external probes and tests its own correction architecture.

Resource efficiency and viability margin

This should replace **Compression** as the headline dimension while preserving compression inside it.

Substrate-neutral definition: the system maintains its useful distinctions and adaptive capacity at a sustainable cost, with enough reserve to survive perturbation and fund future updating.

MDL supplies one component—parsimonious representation—but allostasis and engineering health add what “compression” lacks: **the cost of maintaining adaptive regulation and the reserve before failure.**

⁵⁵ A tiny model that discards rare but consequential distinctions is not healthier merely because it is cheap.

For a **human**, this cannot ethically mean “healthcare spending per unit IQ.” Better measures include cognitive effort and fatigue under standardized work, recovery after load, task performance per unit time, and reserve under dual-task or stress conditions. Resource need must never be translated into moral worth.

For **AI**, compute energy, latency, memory, data and monetary cost per validated unit of useful performance, plus reserve capacity under peak demand. Track whether efficiency gains disproportionately destroy tail performance or calibration.

For an **institution**, measure staff/time/resources consumed per accurate decision or service outcome, backlog, redundancy needed for safety, uncommitted reserve, and the ability to absorb shocks without eliminating monitoring and learning functions.

Levels:

1 — Non-viable: upkeep exceeds available resources or essential function fails under ordinary load.

2 — Brittle: survives nominal conditions but lacks reserve; efficiency depends on lost distinctions or deferred maintenance.

3 — Sustainable: ordinary operation and learning are affordable.

4 — Slack-positive: significant shocks can be absorbed while maintaining learning and correction.

5 — Regenerative: efficiency improvements preserve fidelity and release resources that demonstrably expand future adaptive capacity.

Network reciprocity and commons effect

This is a more cautious version of **Contribution**.

Substrate-neutral definition: the system’s participation changes the future quality, diversity and availability of the informational and material environment on which it and other nodes depend.

It is explicitly a **node-network relation**, not a measure of moral goodness and not yet proven to be an intrinsic component of intelligence.

For a **human**, indicators might include production of reusable knowledge, care, mentoring or error correction, but measurement must also capture extraction, misinformation and uncompensated burdens imposed on others. Raw productivity would be a bad proxy.

For **AI**, assess provenance-respecting use of source material, whether outputs increase or pollute future information quality, induced human verification burden, concentration/dependency effects, and whether

feedback or discoveries are returned to maintainers or public knowledge systems. Recursive-data results show one concrete way an informational commons can deteriorate. ⁵⁶

For an **institution**, measure public-goods production, externalized costs, knowledge sharing, independent capacity maintained outside the organization, resource depletion and the quality of governance around shared infrastructure. Ostrom’s framework shows why sustainable commons require more than simply counting present output. ⁵⁷

Levels:

- 1 — Depleting:** systematically reduces the commons or destroys independent corrective capacity.
- 2 — Net extracting:** benefits mainly by consuming stocks generated elsewhere.
- 3 — Neutral/replenishing:** use is approximately matched by replenishment and external costs are controlled.
- 4 — Contributing:** measurably improves shared informational/resource capacity.
- 5 — Commons-generative:** its participation increases both the stock and other nodes’ capacity to generate, verify and maintain it.

The commonality lies in the **questions**, not necessarily the raw units. “Brier score” cannot be applied to a care institution exactly as to a classifier, just as blood pressure and an engineering vibration spectrum are not interchangeable. Substrate neutrality should mean *construct invariance*, not identical sensors.

A serious validation programme would therefore need to demonstrate measurement invariance at the abstract level: analogous perturbations should move analogous dimensions, dimensions should have predicted relationships with future failure, and the instrument should distinguish systems known to fail for different mechanisms. Until that work is done, the five dimensions are a **research ontology**, not a validated scale.

Combining and valuing

The literature argues strongly against jumping directly to one “QAI” number.

Profile first, index second

HELM’s central methodological choice—to expose multiple metrics and their trade-offs rather than obscure them in one accuracy number—is exactly right for an early intelligence-health instrument. ⁵⁸ The first output should therefore be a five-element profile, accompanied by context, uncertainty and support condition:

$$IHP = (A, E, C, V, N)$$

where A is adaptive learning/retention, E epistemic fidelity, C correctability, V resource viability, and N network reciprocity.

A second layer could report **red-line failures**: for example, no high-stakes deployment could receive an overall “healthy” designation while epistemic fidelity or correctability is at level 1. That is different from mathematically multiplying ordinal ratings.

The product hypothesis is testable, but raw multiplication is not justified

The product idea becomes principled only under a specific causal interpretation. Suppose long-run staying-in-step requires a sequence:

$$P(\text{remain viable}) = P(\text{detect change}) \times P(\text{learn} \mid \text{detected}) \times P(\text{retain} \mid \text{learned}) \times \dots$$

Then a product is mathematically meaningful because its factors are **conditional probabilities in a causal chain**. A zero really does close the path.

But multiplying five 1–5 questionnaire ratings has no corresponding justification. It creates arbitrary ratios, assumes a common zero, and double-counts correlated constructs. The safer progression is:

profile → empirical interaction model → validated composite if useful.

A geometric mean could be used descriptively to penalize very low dimensions, but should be labelled an index convention rather than a theorem.

There is another important qualification from EbE's own CORE: feedback can be costly or misleading, and a sealed configuration may do perfectly well in a stable niche. Therefore even “zero Correction means zero health” cannot be an unconditional truth. It is a prediction specifically for environments in which consequential unanticipated change is possible over the chosen horizon. That conditionality should be built into the construct definition.

Prediction rather than preference can set statistical weights

QALY valuation asks, roughly, how health states should be valued relative to time. Its long methodological history shows how sensitive elicited numbers can be to behavioural assumptions and elicitation design.

filecite turn0file0 An intelligence-health index does not have to repeat this architecture.

For a **descriptive/predictive construct**, weights could be learned from prospective outcomes: which baseline dimensions predict maintaining calibrated, useful function after shocks, distribution changes and resource constraints? Survival analysis, longitudinal latent-variable models and causal perturbation studies could estimate those relationships.

But empirical prediction cannot answer the normative question “whose errors matter?” A model that predicts organizational survival might reward authoritarian suppression of dissent if that prolongs a regime. A system optimized for its own persistence might externalize catastrophic costs. Thus **empirical weights should determine predictive validity; rights, safety and distributional constraints require separate normative governance**.

Individual versus supported intelligence should be central, not ancillary

The best analogue is ICF plus cognitive reserve, not a deficit correction. ⁵⁹ For every system, record at least two conditions:

$$IHP_{\text{intrinsic/minimal support}} \quad \text{and} \quad IHP_{\text{supported/ecological}}$$

and a dimension-specific support effect:

$$\Delta_{\text{support}} = IHP_{\text{supported}} - IHP_{\text{minimal}}$$

The gap is **not a penalty**. It quantifies how much effective intelligence is constituted or restored by the network.

For a person with dementia, calendars, carers, medication systems, familiar environments and durable records may preserve epistemic and practical functioning. Removing those aids merely to obtain a “pure” individual score would be both scientifically artificial and, prospectively, unethical. When direct withdrawal is harmful, support effects should be estimated through benign crossover tasks, natural experiments, assistive-technology introduction or causal modelling rather than deprivation.

Exactly the same distinction applies to AI. A frozen base model may be poor at current factual knowledge but excellent when coupled to retrieval and tools. An institution may depend heavily on external auditors or scientific advisory networks. Calling those supports “not really part of its intelligence” is a definitional choice, not an empirical discovery.

Equity is the hardest problem

QALYs have long faced the objection that some implementations can disadvantage people with disabilities when baseline health limits attainable QALY gains or when general-public valuations undervalue life with disability; the literature contains both serious discrimination critiques and arguments that discrimination is not an inevitable property of every QALY-based allocation rule. ⁶⁰

An intelligence-health score creates an even more dangerous possibility: attaching an apparently objective numerical value to cognitive impairment. That would be unacceptable. **The score must never represent the worth of a person, entitlement to resources, moral agency, or value of a life.** It describes a functional relationship among a particular system, challenge set, support environment and time horizon.

The individual/supported distinction can reduce rather than worsen discrimination if designed correctly: it makes the effect of environmental accommodation visible. A low unaided score with a high supported score becomes evidence that **support works**, not that the person is worth less. This turns the metric toward a policy question familiar from ICF and capability theory: *what environment enables this person to remain connected, active and epistemically supported?* ⁶¹

What this teaches the meta-model

The literature strengthens several parts of EbE, weakens others, and suggests one important reorganization.

Strengthened: the idea that health is dynamic rather than a fixed optimal state has substantial independent support from Canguilhem, allostasis, resilience, viability theory and Positive Health. ⁶² The importance of external correction and independent variation is also independently supported by network epistemology, organizational learning and AI failure modes. ⁶³ The supported-intelligence claim is well aligned with ICF, cognitive reserve and distributed cognition. ⁶⁴ Finally, resource cost belongs inside the

theory: allostatic load, engineering health and AI efficiency all show that adaptation cannot be evaluated without what it costs. ⁶⁵

Challenged: “compression” is not yet demonstrated as an independent health dimension; it is better treated as one route to resource viability. “Coherence” should not mean consistency but **calibrated correspondence with evidence**. “Contribution” is plausibly a determinant of network health but not yet a demonstrated component of individual intelligence health. And the claim that no intelligence can “certify itself from the inside” should be presented carefully: internal confidence cannot establish external truth, but agents can sometimes prove conditional propositions from internally available axioms or data. The substantive thesis concerns **empirical world-model validity and the impossibility of making one’s own felt certainty an independent external validator**, which is stronger when stated narrowly.

Missing: the existing five dimensions understate the stability-plasticity trade-off. Retention must be explicitly part of Change. They also need a separate distinction between **mechanism and outcome**. “Remaining in step with the world” is best treated as the criterion outcome against which dimensions are validated, rather than as another dimension. This prevents circularity: one measures learning, fidelity, correction, viability and network effect at t_0 , then asks whether those measures actually predict successful adaptation at t_1 .

The meta-model would become clearer if expressed at three nested levels:

Agent mechanisms → Agent–network interfaces → long-run viability

Adaptive learning, epistemic fidelity and resource viability are primarily agent/system mechanisms; correction and reciprocity are inherently relational; persistence or “staying in step” is the longitudinal outcome. This is more precise than treating all five as ontologically equivalent properties.

Testable predictions

Independent correction prediction. Systems with at least one correction route whose errors are imperfectly correlated with the focal system should recover from novel false beliefs more reliably than equally connected systems whose sources derive from the same underlying model or data. Zollman’s models already motivate such an effect. ⁶⁶ A decisive test could compare human teams, multi-agent AI systems and institutional panels under experimentally induced shared versus independent evidence sources. **Against EbE:** after controlling for source competence and cost, independence provides no advantage under distribution shift, or highly homogenized systems correct just as reliably.

Supported-intelligence prediction. Ecologically supported IHP should predict real-world functioning and error recovery better than unaided cognitive performance alone, especially where memory or executive impairment is present. Cognitive reserve and ICF make this plausible. ⁶⁷ **Against EbE:** support configuration adds no prospective explanatory power once individual cognitive measures and disease severity are known.

Stability-plasticity prediction. High learning rate without retention should produce worse long-run staying-in-step than a balanced adaptation-and-retention score, across human sequential learning, continual AI training and organizational policy revision. Catastrophic forgetting makes this particularly

testable in AI. ²⁰ **Against EbE:** current-task adaptation alone predicts long-run performance just as well; retained prior competence adds no information.

Bottleneck/product prediction. Models with interaction terms or conditional bottlenecks should predict catastrophic failure better than an additive score: e.g., excellent learning and efficiency will not rescue a system whose consequential errors are systematically undiscoverable. **Against EbE:** additive models consistently outperform bottleneck/product models out of sample and near-zero correctability has no disproportionate effect.

Commons-level prediction. Network reciprocity should predict future quality and diversity of the shared information environment even after controlling for participants' individual intelligence. Recursive-data degradation and commons theory provide candidate mechanisms. ⁶⁸ **Against EbE:** contribution/extraction has no independent relationship with future network epistemic quality or viability. Such a result would be especially informative: it would imply that Contribution belongs in ethics/governance rather than in a theory of intelligence health.

These predictions are more important for EbE than conceptual resemblance. The literature can show that EbE has many distinguished relatives; only prospective cross-domain tests can show whether its proposed abstraction buys explanatory power.

Open questions and risks

The first danger is **reification**. A five-dimensional profile would be a model of selected functions, not a meter detecting an independently existing substance called intelligence health. Psychometrics, collective-intelligence research and AI benchmarks all show that a construct can acquire a clean score long before its ontological interpretation is settled. ⁶⁹

Second is **cross-substrate false equivalence**. A human updates through biological learning embedded in needs, embodiment and social history; a frozen language model may not update at all during ordinary inference; an institution changes through people, records, incentives and rules. Calling all three "learning" can be scientifically useful only if the common definition generates successful cross-domain predictions. Otherwise it is analogy.

Third is **anthropomorphizing AI**. Functional degradation, calibration, drift and remaining useful service life can legitimately be called system health in an engineering sense. ⁷⁰ None of that establishes suffering, consciousness, interests or moral patienthood. Those require separate arguments and evidence. ²⁶ Conversely, future evidence of machine consciousness would create welfare obligations that a functional IHP could not capture.

Fourth is **medicalizing disagreement in humans**. Low "correctability" must never mean disagreement with authority, majority opinion or clinical advice. The sycophancy literature gives the cleanest warning: agreement can increase while truthfulness decreases. ²³ Correctability must therefore be measured against independently checkable evidence and the ability to distinguish valid from invalid correction.

Fifth is **political misuse**. Governments, employers, insurers or schools could transform a contextual measure into a rank ordering of persons. Disability history gives ample reason to anticipate this. Any human

application would need explicit prohibitions against interpreting IHP as person-worth, civic competence or resource entitlement, strong governance over who chooses challenge sets, and routine reporting of supported alongside unsupported functioning.

Sixth is **gaming and Goodhart effects**. Once institutions are rewarded for “correction events,” they can generate superficial corrections; if AI models are rewarded for verbal uncertainty, they can learn performative hedging; if commons contribution is measured by quantity of output, systems can flood the commons. The measure therefore requires hidden or changing validation tasks, outcome-based audit, and periodic redesign.

Seventh is **timescale dependence**. A high-throughput institution may look extremely efficient over a year while hollowing out expertise needed ten years later; an AI update may improve today’s benchmark while degrading rare capabilities; adaptation that is beneficial over hours can constitute allostatic overload over years. All scores therefore require a declared horizon.

Finally, the most important unresolved question is **what “staying in step” ultimately means**. Survival is too weak: false institutions, parasitic organizations and maladaptive ideologies can persist for long periods. Accuracy is too narrow: perfectly accurate systems may be too costly to maintain. Goal achievement is goal-dependent and can reward harmful goals. The strongest candidate criterion emerging from the combined literatures is therefore:

A healthy intelligence is a system that can maintain warranted, useful contact with a changing reality, revise that contact without destructive loss, keep consequential errors discoverable, and sustain the resources and network relations that make continued correction possible.

That formulation is a **synthesis**, not an established definition. Its value is precisely that its pieces can be operationalized and then allowed to fail.

Annotated bibliography

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The resulting research programme is therefore narrower—and, in my assessment, stronger—than “invent a QALY for intelligence.” The immediate scientific object should be a **substrate-neutral profile of adaptive epistemic viability**, validated prospectively and reported at both intrinsic and supported levels. The crucial empirical question is not whether humans, AI systems and institutions can all be forced onto the same five-point questionnaire. It is whether the **same causal architecture predicts when each will remain corrigibly connected to a changing world**.

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